3d Computer Vision

§3 Data Acquisition

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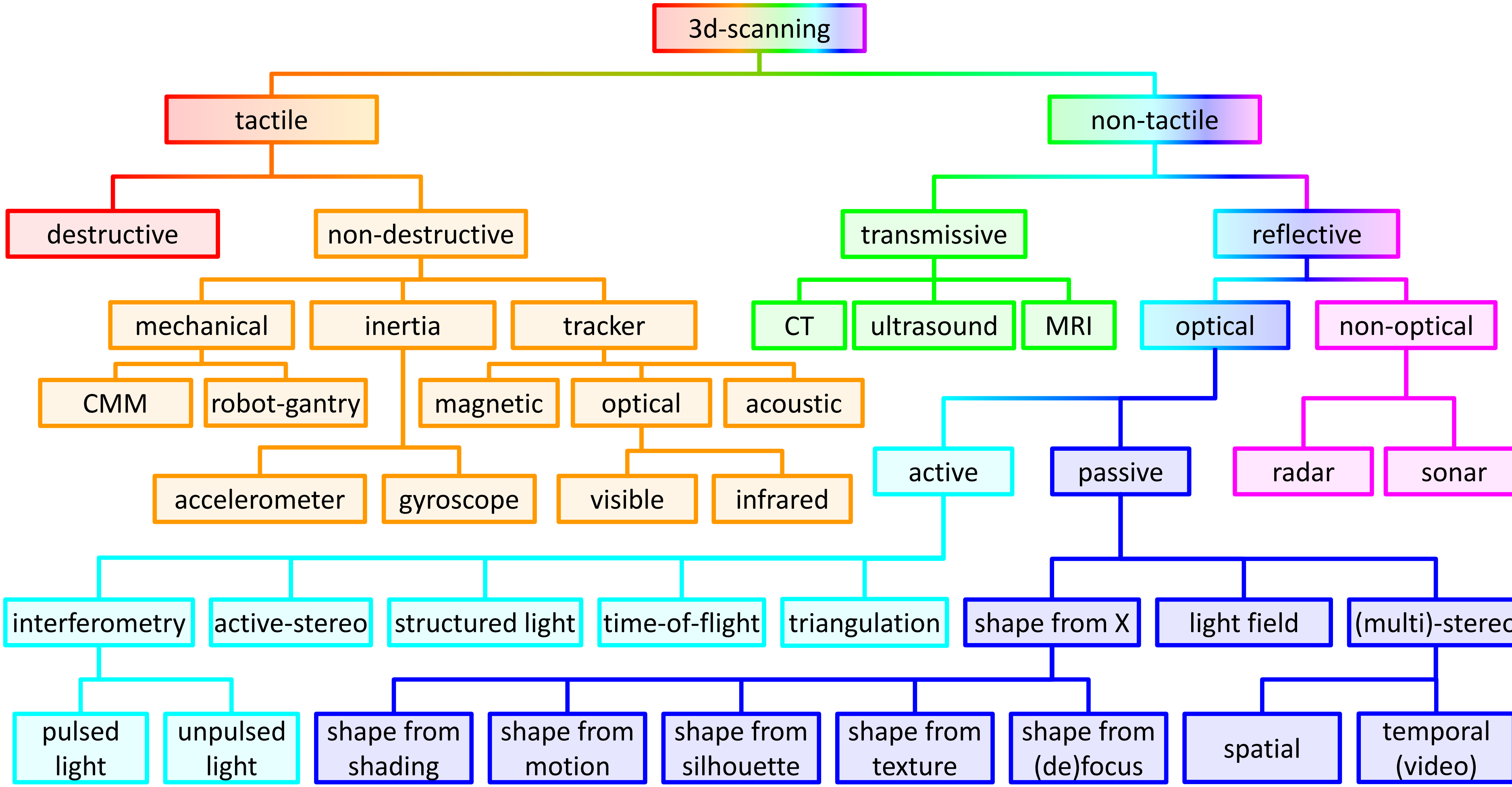
Content

1. Overview
2. 3d data acquisition examples
3. Stereo vision
4. Laser scanning

1. Overview



3d-scanning
tactile
non-tactile
destructive
non-destructive
transmissive
reflective
mechanical
inertia
tracker
CT
ultrasound
MRI
optical
non-optical
CMM
robot-gantry
magnetic
optical
acoustic
accelerometer
gyroscope
visible
infrared
active
passive
radar
sonar
interferometry
active-stereo
structured light
time-of-flight
triangulation
shape from X
light field
(multi)-stereo
pulsed light
unpulsed light
shape from shading
shape from motion
shape from silhouette
shape from texture
shape from (de)focus
spatial
temporal (video)

Overview

10^{-10}

10^{-6}

10^{-2}

[m]

- Tunnel effect microscopy: < nm
- Atomic force microscopy: < nm
- Interferometry: nm
- Confocal scanning: µm
- Depth from focus: µm
- Laser line triangulation: µm − mm
- Structured light: µm − mm
- Stereo triangulation: µm − cm
- MRI/CT: µm − cm
- Time of flight (ToF), Lidar: cm
- Radar: cm − m

2. 3d-Data Acquisition

Often two stage approach:

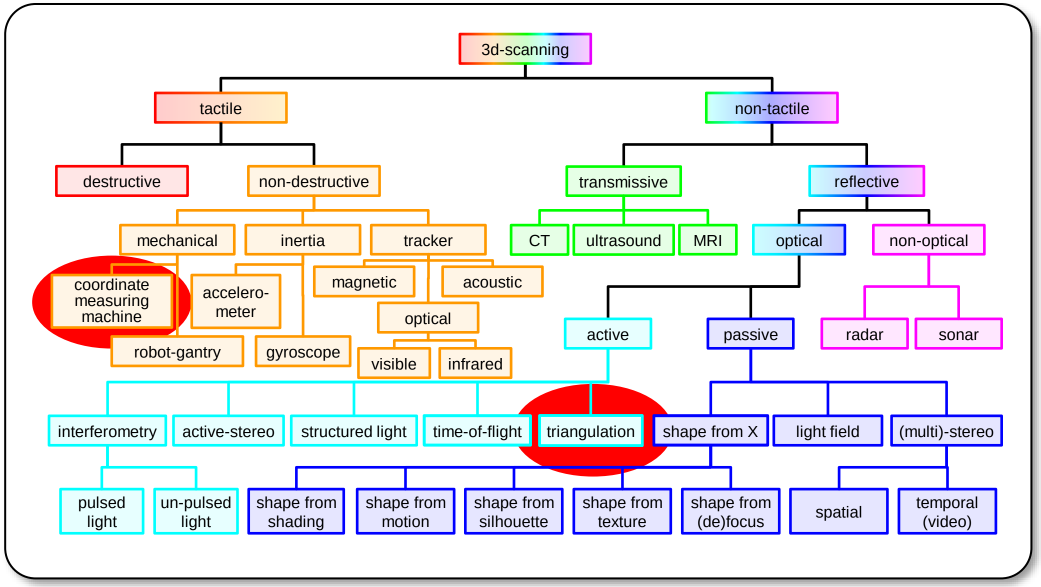
1. Localization of the sensor.
2. Localization of 3d data relative to the sensor.

Examples

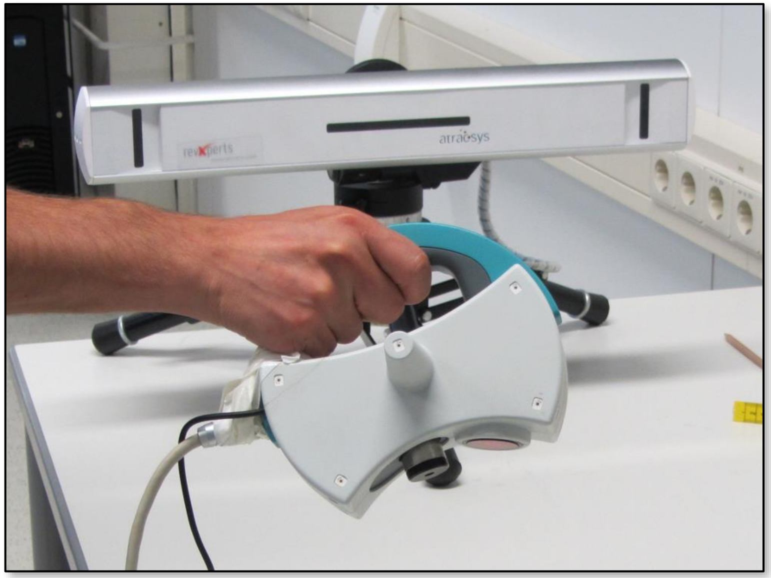
FARO

Combination of

- CMM and
- laser triangulation



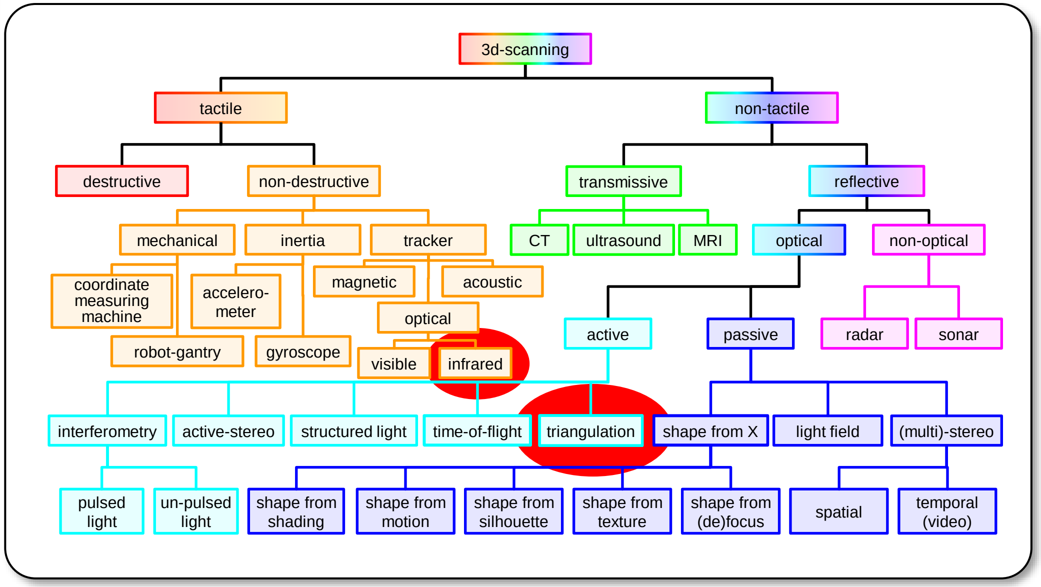
Examples



arrac sys

Combination of

- infrared tracker and
- laser triangulation



Examples

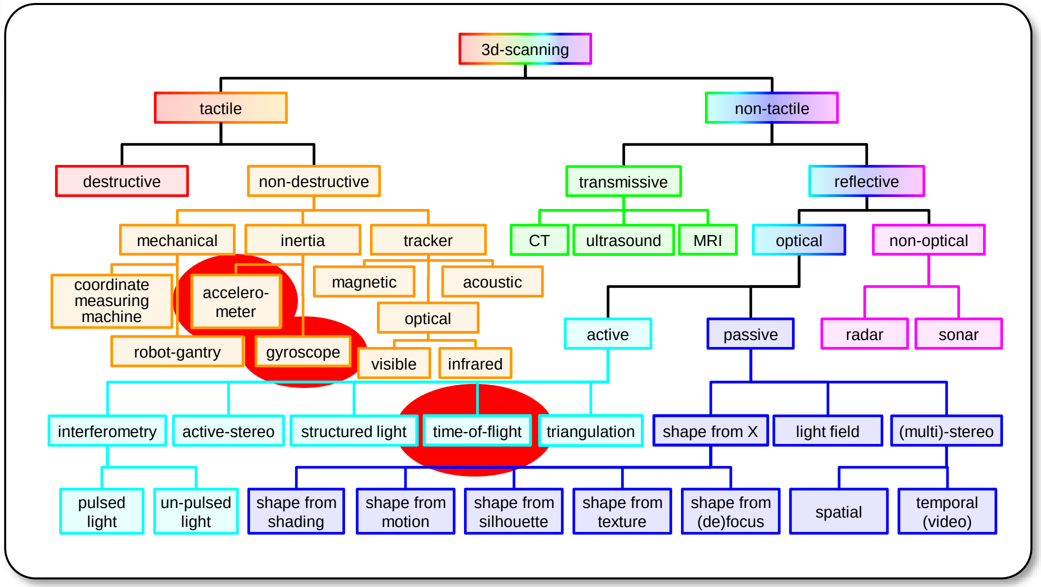
Microsoft HoloLens

Extras:

- 2 RGB video
- 4 microphones
- ambient light sensor

Combination of

- accelerometer, gyroscope, magnetometer,
- two time-of-flight cameras

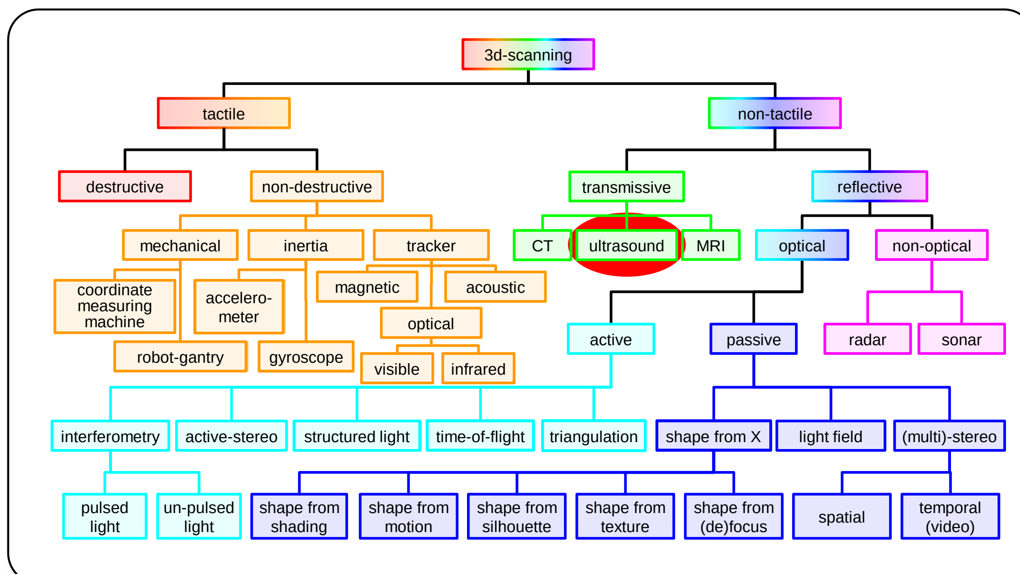


2. 3d-Data Acquisition

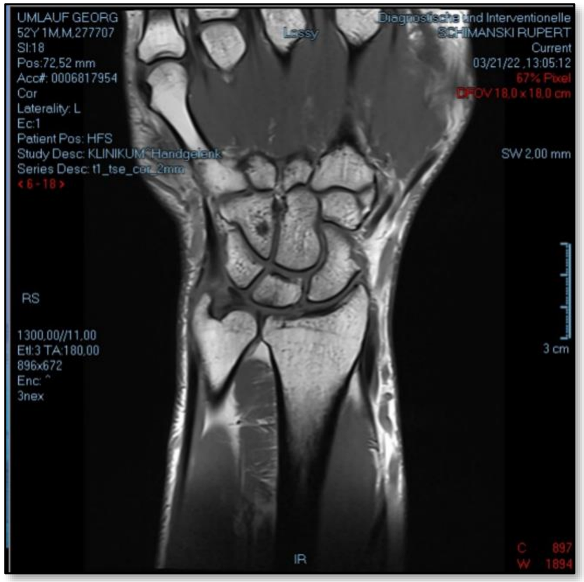
Courtesy: Ulla + Emmy B.

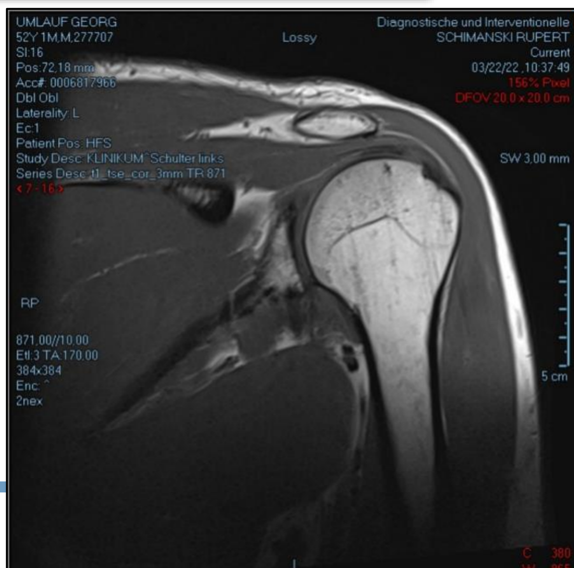
Combination of

- multiple ultrasound measurements.
- Analog for CT, MRI, etc.

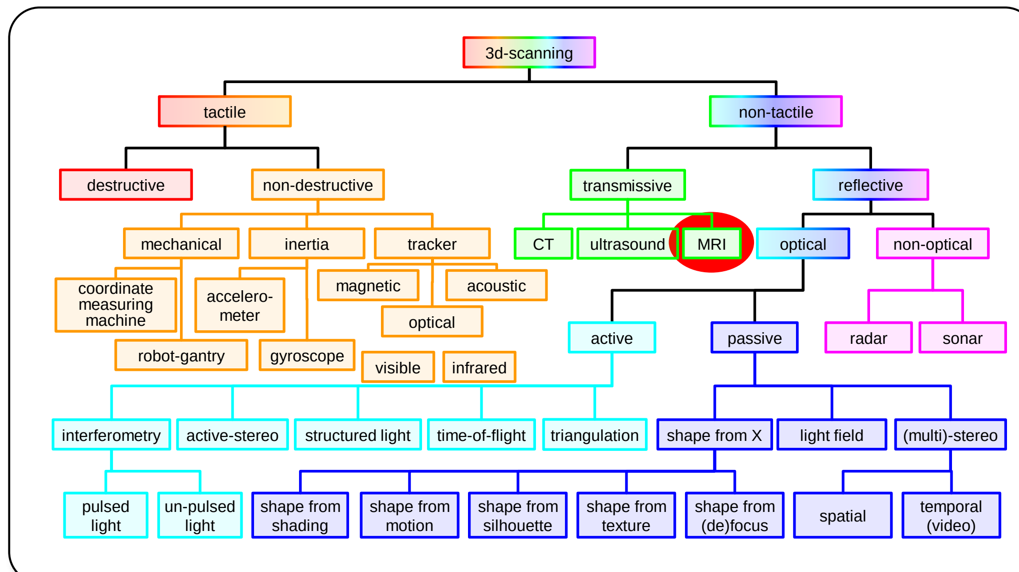


2. 3d-Data Acquisition

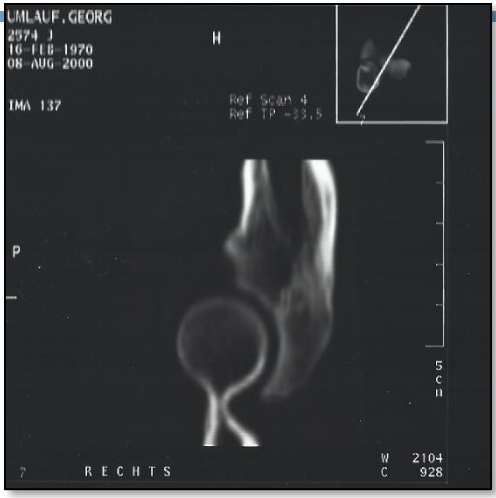




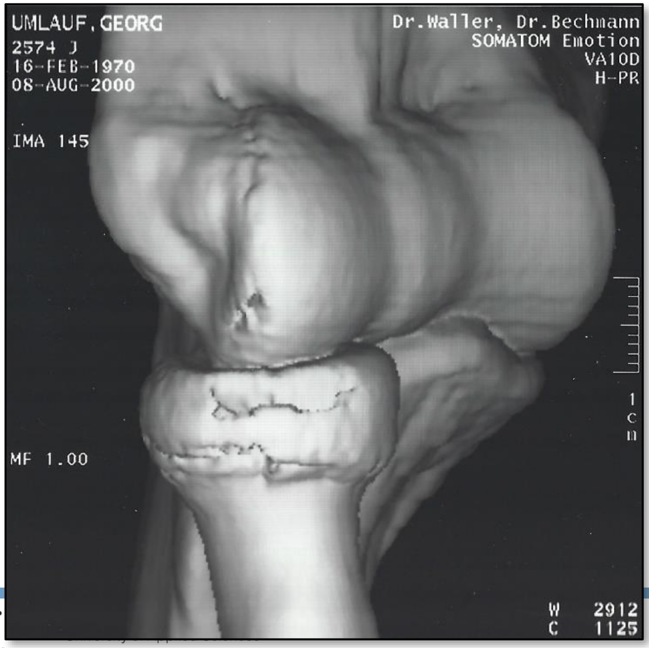
Wikipedia

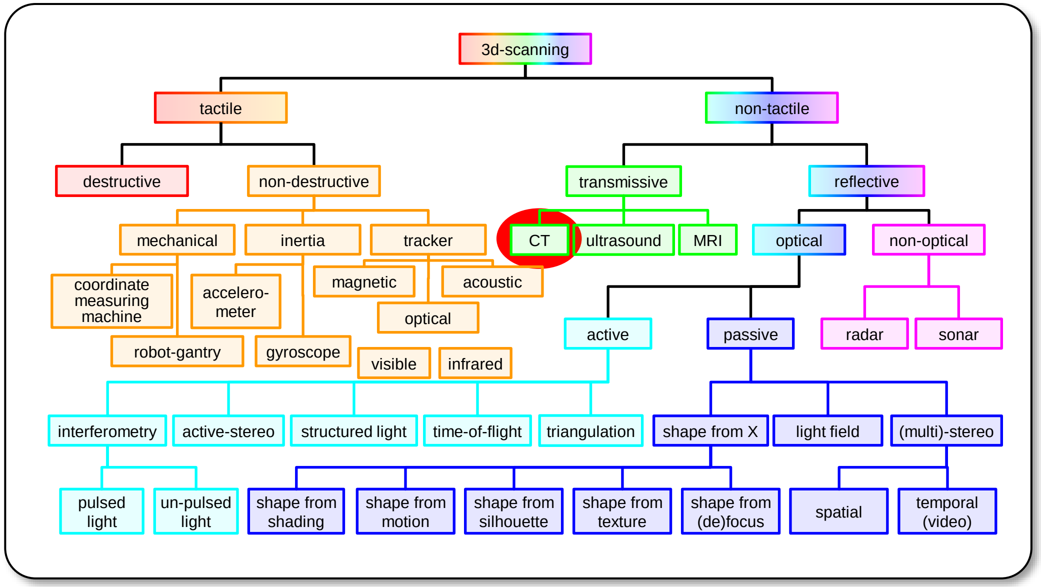


2. 3d-Data Acquisition

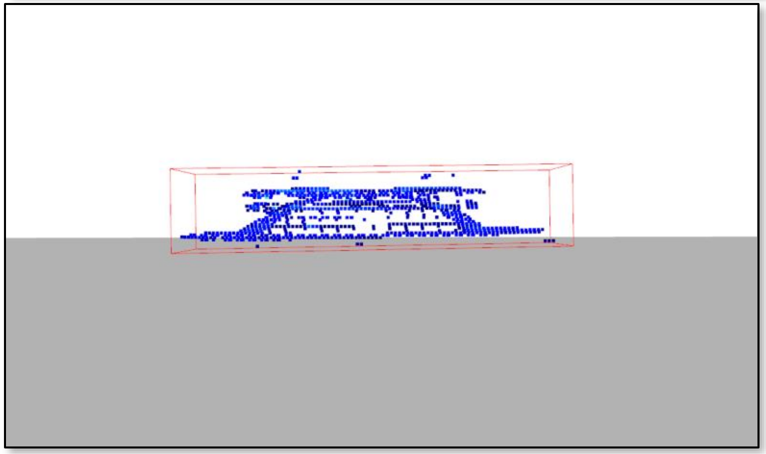


Wikipedia





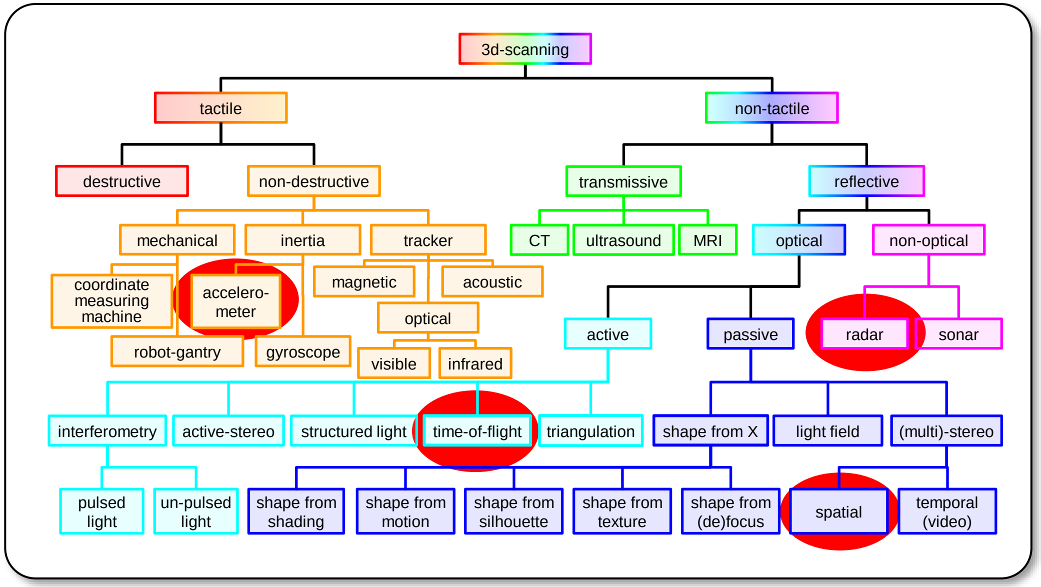
Examples



IOS@HTWG

Combination of

- two stereo cameras
- LIDAR
- IMU + GPS
- radar (research)



Content

§3.1 Basics of stereo vision

Stereoscopic vision and depth perception is based on

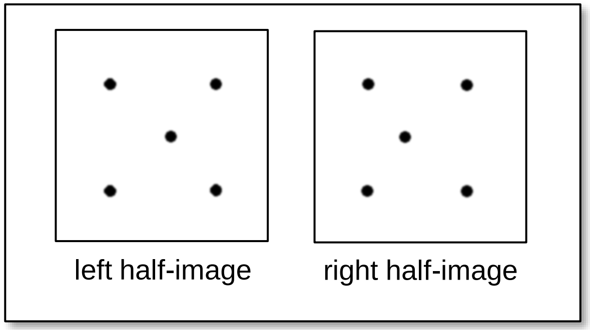
- differences (**disparities**) between two images from different perspectives (e.g., left/right eye) and
- the resulting parallax.

- **Parallax:** change of position of an object, as seen from different perspectives,
  - different image coordinates from identical world coordinates of an object,
  - different camera coordinates from identical world coordinates of an object.
- Both images together form the so-called **stereogram**, the individual images are so-called **half-images**.

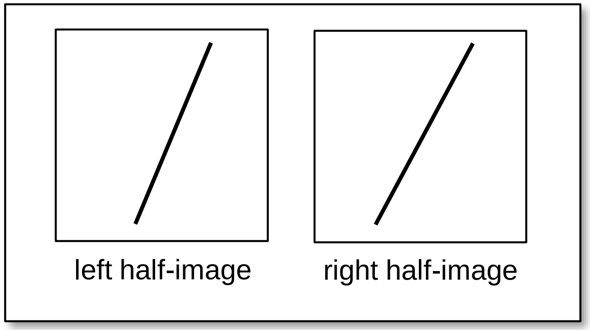
Disparity (1)

Differences in two (or more) images of the same object.

(1) Point disparity – horizontal and vertical disparity
- horizontal disparity, e.g., in the normal case (see 3.2),
- vertical disparity, e.g., when the viewing directions are not perfectly co-planar.

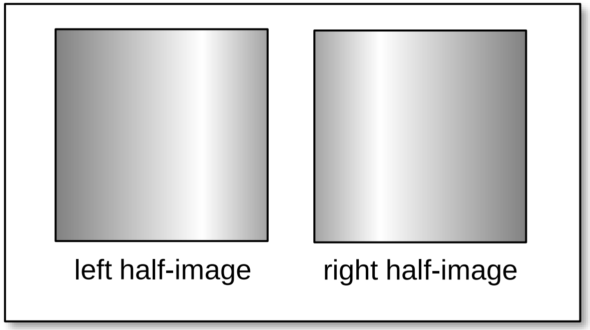


(2) Orientation disparity rsp. general form and size differences between both images
- Leads always also to point disparities.



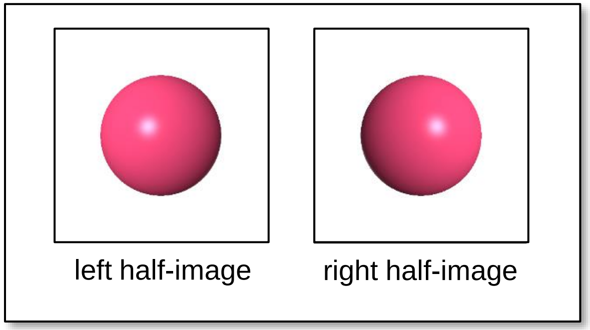
Disparity (2)

(3) Gray value disparity and shading disparity caused by different normals of object points in diffuse reflection in the images.



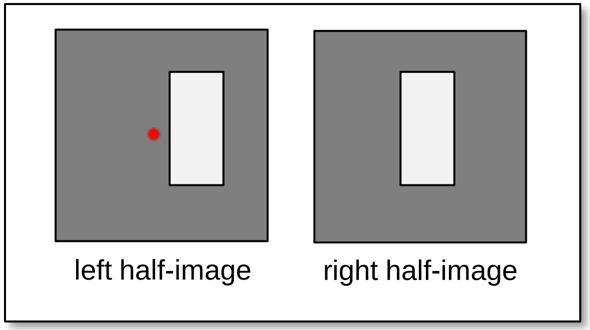
(4) Photometric disparity caused by surface reflectivity, aka specular reflection

- Highlights occur at different image positions.



(5) Monocular occlusion at object edges

- Difficult for stereo algorithms!



§3.1 Basics of stereo vision

Crucial for the reconstruction is the information about the relative position and orientation of both cameras:

- The extrinsic camera parameters differ in camera direction/orientation, center of projection.
- The intrinsic camera parameters are (usually) identical: focal length, image principal point.

Remark: The camera calibration must be very accurate, because small errors in the calibration lead to large errors in the reconstruction.

Stereoscopic geometry

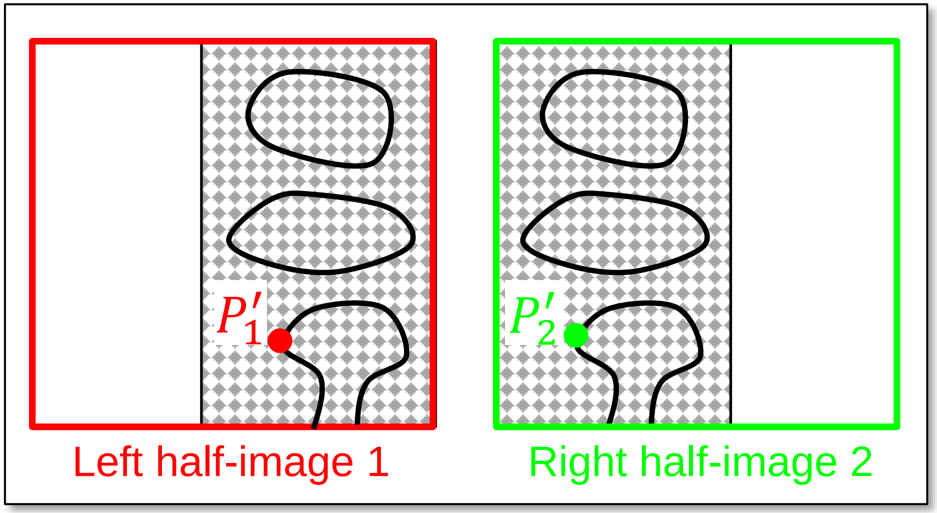
- Parallel camera axes, i.e., so-called normal case.
- Converging/diverging camera axes.

Content

3.2 Normal case of stereo vision

- **Input:** Two corresponding image points P_1', P_2' of the same object point P, from two half-images that overlap ca. 60%.
- **Output:** Reconstruct the object from corresponding image points P_1', P_2'
- **Assumptions**
  - intrinsic camera parameters f, H,
  - extrinsic camera parameters R, N and
  - relative positions of cameras

  are known.
- **Notation:** Index 1 denotes quantities in the first/left half-image, index 2 denotes quantities in the second/right half-image.



3.2 Normal case of stereo vision

- **Normal case:** The reconstruction of a 3d object becomes particularly simple, if (for $N_i = O_i, H_i = O_i', i = 1,2$) both view directions
  - are perpendicular to the base line in x-direction, i.e., the line connecting the centers of projection,
  - are at distance b (aka base length), and
  - are parallel to each other.

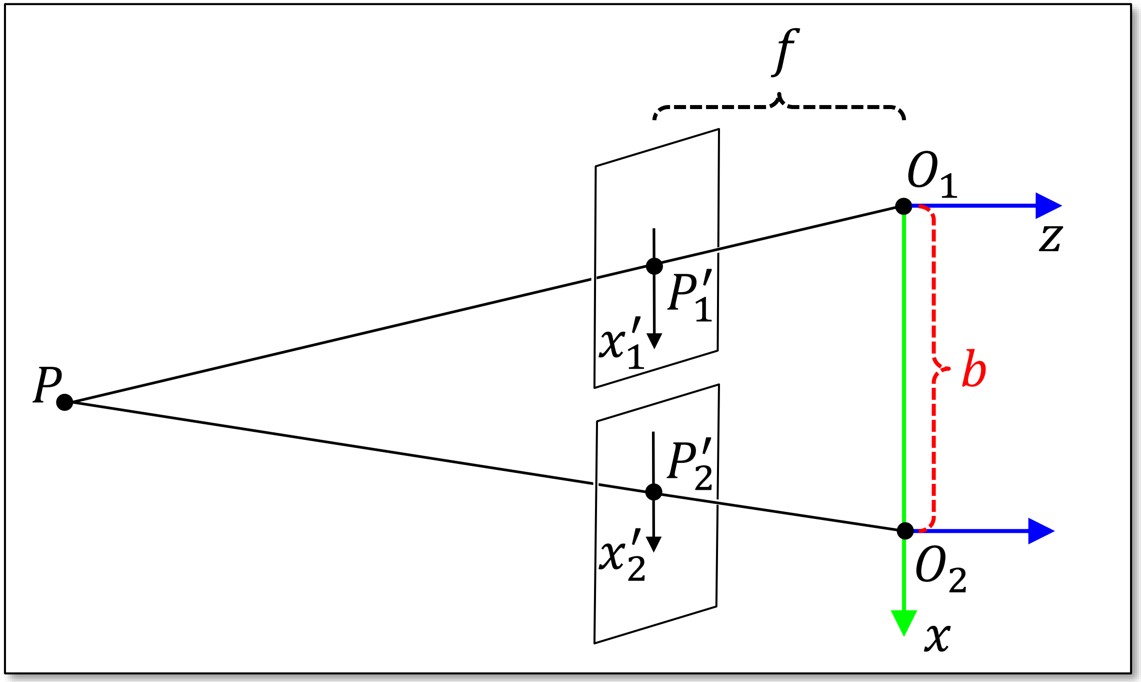
▶ $N_1 = (0,0,0)^t$,

$N_2 = (b, 0,0)^t$,

$H_1 = H_2 = (0,0)^t$,

$R_1 = R_2 = I$,

$f_1 = f_2 = f$.



3.2 Normal case of stereo vision

- **Spatial forward step:** The general spatial reconstruction

$$x = x_0 + (z - z_0)\frac{r_{11}(x'-x'_0)+r_{12}(y'-y'_0)-r_{13}f}{r_{31}(x'-x'_0)+r_{32}(y'-y'_0)-r_{33}f}$$

$$y = y_0 + (z - z_0)\frac{r_{21}(x'-x'_0)+r_{22}(y'-y'_0)-r_{23}f}{r_{31}(x'-x'_0)+r_{32}(y'-y'_0)-r_{33}f}$$

simplifies in the normal case to

left image: $x = -z\frac{x'_1}{f}, \quad y = -z\frac{y'_1}{f}$

right image: $x = b - z\frac{x'_2}{f}, \quad y = -z\frac{y'_2}{f}$

Normal case:

- $N_1 = (0,0,0)^t$
- $N_2 = (b,0,0)^t$
- $H_1 = H_2 = (0,0)^t$
- $R_1 = R_2 = I$
- $f_1 = f_2 = f$

- The space point $P = (x,y,z)^t$ corresponds to two image points $P'_i = (x'_i, y'_i)^t, i = 1{,}2$, with the same y'-coordinate.
  - there is no y'-parallax,
  - there is only a x'-parallax $p_{x'} = x'_1 - x'_2$.

3.2 Normal case of stereo vision

- The correspondence of the image of a point in the left half-image is in the same row in the right half-image.
- This simplifies the search for the correspondences.
- Equating the x-coordinates from both images, yields

$$-z\frac{x_1'}{f} = b - z\frac{x_2'}{f}$$

and thus

  - $z = -f \cdot \frac{b}{x_1' - x_2'} = -f\frac{b}{p_{x'}}$, ($p_{x'} = x_1' - x_2'$ is the x'-parallax)
  - $y = -z\frac{y_1'}{f} = -z\frac{y_2'}{f}$ (for control)
  - $x = -z\frac{x_1'}{f}$.
- The larger the parallax, the closer is the point to the camera.

Content

3.3 Error propagation

- In stereo reconstruction the spatial positions are computed indirectly from the 2d-image positions x_1', y_1' and the parallax $p_{x'}$.
- ▶ Errors in the 2d-image positions cause errors in the computed space positions.
- ▶ How does the error in the 2d-image positions propagate to the space positions?

3.3 Error propagation

- How does the error in the 2d-image positions propagate to the space positions?
- *Gauss error propagation:* The variance σ_v of a quantity $v = F(u_1, \dots, u_n)$, computed from defective parameters $u_1, \dots, u_n$, is given by

$$\sigma_v^2 = \sum_{i=1}^{n} \left(\left.\frac{\partial F}{\partial u_i}\right|_{u_1,\dots,u_n} \cdot \sigma_{u_i} \right)^2$$

$$= \left(\left.\frac{\partial F}{\partial u_1}\right|_{u_1,\dots,u_n} \cdot \sigma_{u_1} \right)^2 + \left(\left.\frac{\partial F}{\partial u_2}\right|_{u_1,\dots,u_n} \cdot \sigma_{u_2} \right)^2 + \cdots + \left(\left.\frac{\partial F}{\partial u_n}\right|_{u_1,\dots,u_n} \cdot \sigma_{u_n} \right)^2$$

- **Remark**: This is a good approximation only for sufficiently small errors.

3.3 Error propagation

- **Assumption:** Base length b und principal distance f are exact.
- Apply error propagation to $z = -\frac{f \cdot b}{p_{x'}}, y = -z\frac{y_1'}{f}, x = -z\frac{x_1'}{f}$.
- $\sigma_z^2 = \left(\left.\frac{\partial z}{\partial p_{x'}}\right|_{p_{x'}} \cdot \sigma_{p_{x'}} \right)^2 = \left(\frac{f \cdot b}{p_{x'}^2} \right)^2 \sigma_{p_{x'}}^2$
- $\sigma_z = \frac{z^2}{f \cdot b} \sigma_{p_{x'}}$
- $\sigma_y^2 = \left(\left.\frac{\partial y}{\partial z}\right|_{z,y_1'} \cdot \sigma_z \right)^2 + \left(\left.\frac{\partial y}{\partial y_1'}\right|_{z,y_1'} \cdot \sigma_{y_1'} \right)^2 = \left(\frac{y_1'}{f} \right)^2 \sigma_z^2 + \left(\frac{z}{f} \right)^2 \sigma_{y_1'}^2$
- $\sigma_x^2 = \left(\left.\frac{\partial y}{\partial z}\right|_{z,x_1'} \cdot \sigma_z \right)^2 + \left(\left.\frac{\partial y}{\partial x_1'}\right|_{z,x_1'} \cdot \sigma_{x_1'} \right)^2 = \left(\frac{x_1'}{f} \right)^2 \sigma_z^2 + \left(\frac{z}{f} \right)^2 \sigma_{x_1'}^2$

3.3 Error propagation

- The accuracy of a spatial reconstruction is controlled by two quantities
  - the base ratio $m_V = \frac{b}{z}$ and
  - the image scale $m_B = \frac{z}{f}$.

$$\sigma_z = \frac{z \cdot z}{f \cdot b}\sigma_{p_{x'}} = \frac{m_B}{m_V}\sigma_{p_{x'}}$$

$$\sigma_y = \sqrt{\left(\frac{y_1'}{f}\frac{m_B}{m_V}\sigma_{p_{x'}}\right)^2 + (m_B\sigma_{y'})^2},$$

$$\sigma_x = \sqrt{\left(\frac{x_1'}{f}\frac{m_B}{m_V}\sigma_{p_{x'}}\right)^2 + (m_B\sigma_{x'})^2}.$$

3.3 Error propagation

General

- The error in the z-coordinate grows quadratically with the distance.

Image scale

- The error in all three coordinates grows proportionally to the image scale $m_B = z/f$.

Base ratio

- The error in the z-coordinate grows inverse proportional to the base ratio $m_V = b/z$.
- The errors in the x- and y-coordinates deteriorate hardly in $m_V = b/z$.
- For $m_V = b/z = 1$ all three errors are approximately the same.

Content

3.4 Spatial forward step

- The spatial reconstruction from general spectrograms is called spatial forward step.
  - The half images must overlap in the target point $p = (x, y, z)^t$.

$$x = x_0 + (z - z_0)\frac{r_{11}(x'-x_0')+r_{12}(y'-y_0')-r_{13}f}{r_{31}(x'-x_0')+r_{32}(y'-y_0')-r_{33}f} = x_0 + (z - z_0) \cdot k_x(x', y'),$$

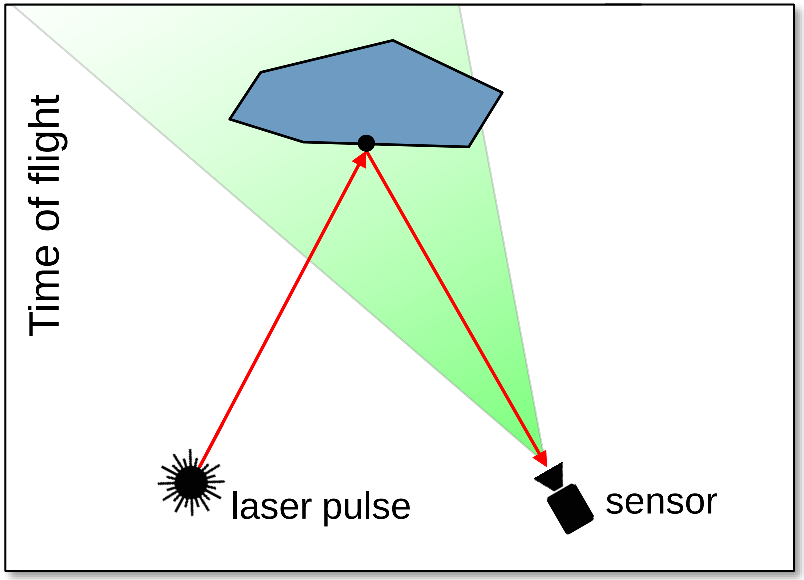
$$y = y_0 + (z - z_0)\frac{r_{21}(x'-x_0')+r_{22}(y'-y_0')-r_{23}f}{r_{31}(x'-x_0')+r_{32}(y'-y_0')-r_{33}f} = y_0 + (z - z_0) \cdot k_y(x', y').$$

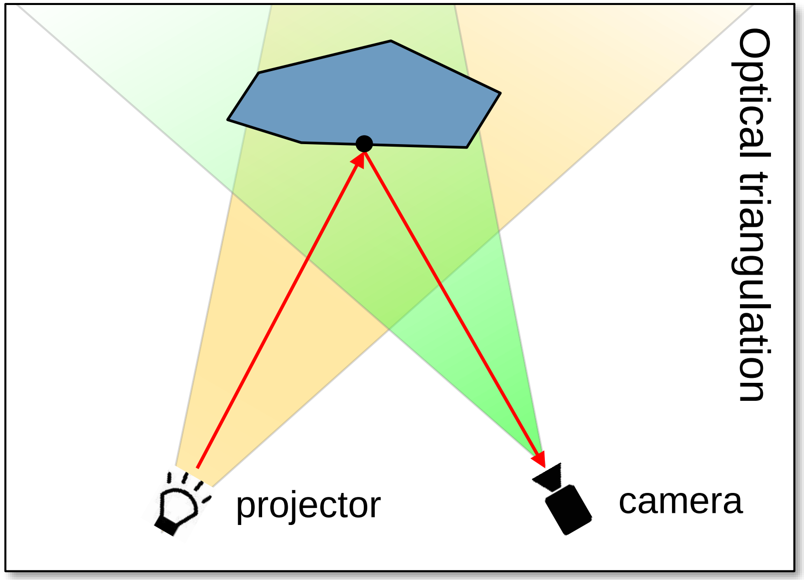
- This yields four equations (in two images) for three unknowns.
- Possible solutions:
  - Ignore the third/fourth equation, solve each system for z, and average both solutions.
    - Relatively inaccurate the larger the deviation from the normal case.
  - **Solution: non-linear regression** …
    - Time consuming computation of correspondences: **epi-polar geometry**…

Content

4. Laser scanning

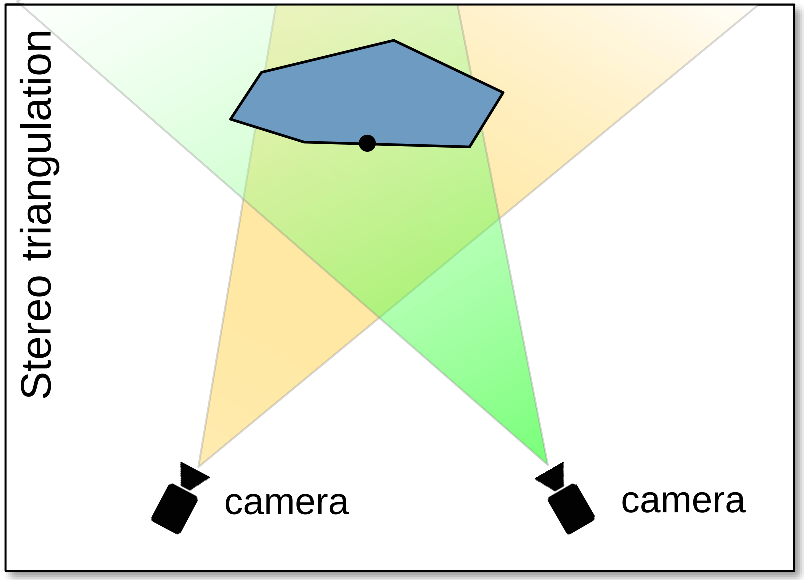
- There are two technical principles of laser scanning:
  - **Time-of-flight (TOF):** Measure the time between a laser pulse at t_0 and its sensor read-back at t_1: $\boldsymbol{d} = \frac{\boldsymbol{t_1 - t_0}}{\boldsymbol{2}}\boldsymbol{c}$ (c is the speed of light)
  - **Optical triangulation:** Measure "distortion" of projected pattern.
    - Measure angle of projected pattern and captured pattern.

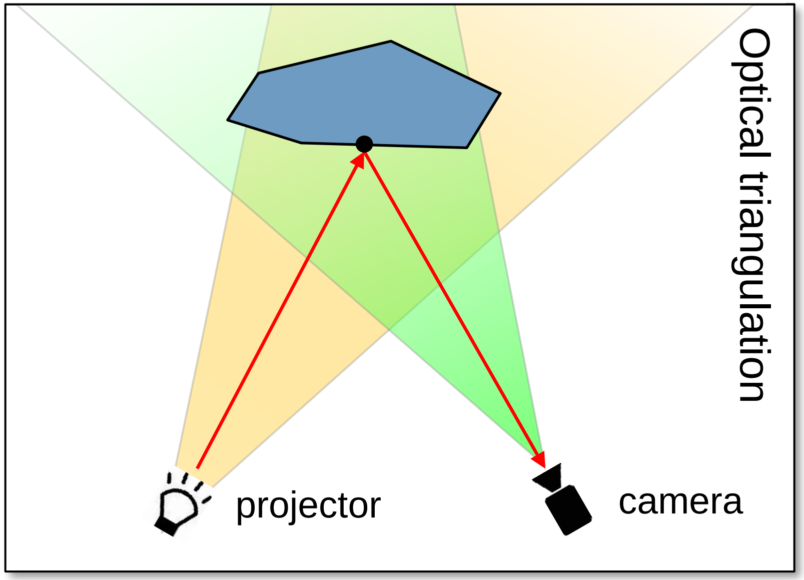




4. Laser scanning

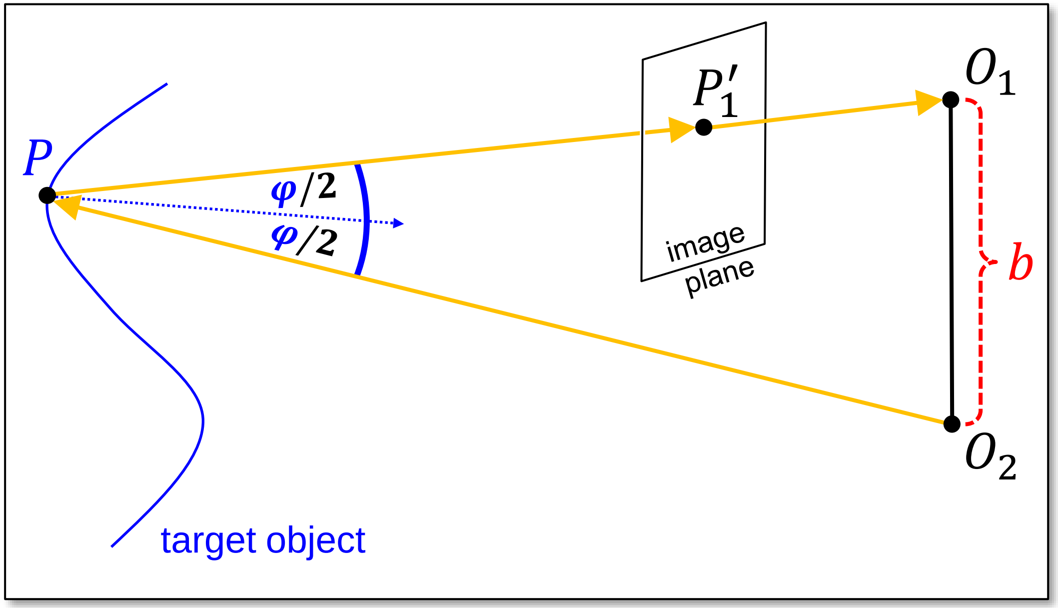
- Difference between stereo vision (left) and triangulation-based laser scanning (right):
  - **Stereo triangulation:** Measure image coordinates of corresponding image points of the same object point.
  - **Optical triangulation:** Measure angle of projected pattern and captured pattern.





Optical triangulation (1)

- One camera (can be more than one…) plus
- projector, e.g., laser point, laser line, structured light, light point pattern, etc.
- Camera in O_1 and projector in O_2 have a fixed relative position (e.g., distance b).
  - ▶ Camera image of the laser point (or pattern) *depends on the geometry* of 3d target object.
  - ▶ Camera image of the laser point (or pattern) *changes with the geometry* of 3d target object.

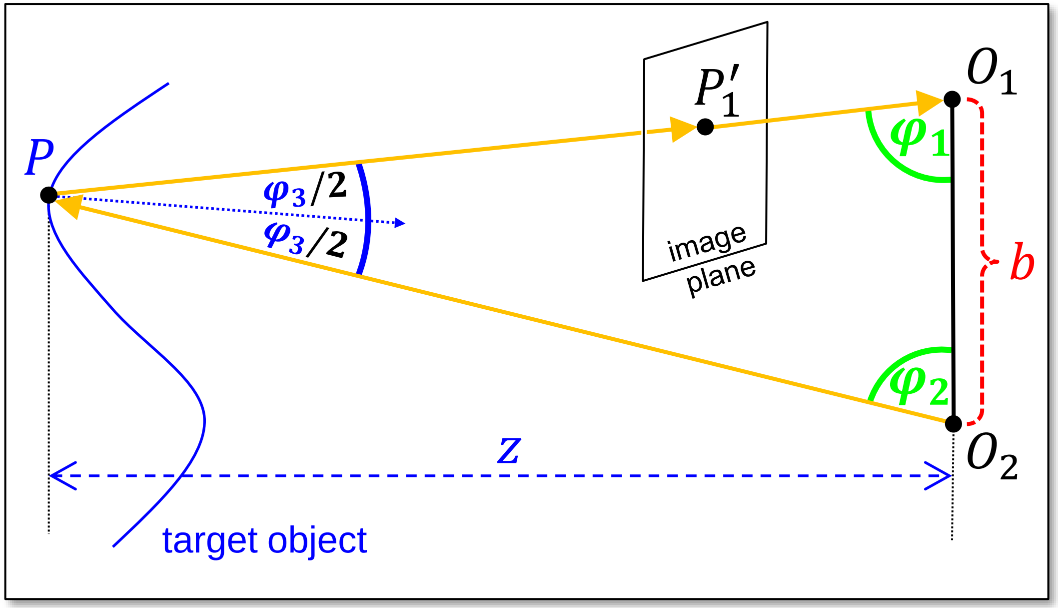


Optical triangulation (2)

- The distance of the object point P is computed via trigonometry (aka triangulation)

$$z = ||P - O_1|| \cdot \sin \varphi_1 = b \cdot \frac{\sin \varphi_1 \sin \varphi_2}{\sin(\pi - \varphi_1 - \varphi_2)}$$

with $\pi = \varphi_1 + \varphi_2 + \varphi_3$.



Attention: Exact calibration is again crucial!

Questions

- Name several 3d data acquisition examples and classify according the lecture’s taxonomy?
- What are disparity and parallax?
- What is the normal case of stereo vision?
- What are the reconstruction equations in the normal case of stereo vision?
- How do errors propagate in the normal case of stereo vision?
- What is the principle of ToF?
- What is the principle of optical triangulation, aka structured light or laser triangulation?